The central Chebyshev-well obstruction: a sharp correction and endpoint counterexamples

Automatic proof packet for arXiv:2201.00410

9 August 2026

Status. Candidate counterexample, likely valid, pending human review. Conjecture 7.4 of Mandich's arXiv:2201.00410v1 is false at the allowed central-well endpoint $j = \kappa/2$, while the packet proves its sharp corrected form for every other well. The same endpoint gives empty-interval counterexamples to Theorems 7.1 and 7.2.

1 Source statements

Mandich studies threshold sequences associated with the Chebyshev polynomial T_κ [1]. In Section 7, Theorems 7.1 and 7.2 assert the existence of decreasing energy sequences for, respectively, odd and even indices

$$1 \le j \le \lfloor \kappa/2 \rfloor,$$

inside the interval

$$\left(\cos^2\frac{j\pi}{\kappa}, \cos\frac{(j-1)\pi}{\kappa}\cos\frac{j\pi}{\kappa}\right).$$

Conjecture 7.4 asks whether, whenever

$$\cos\frac{(j+1)\pi}{\kappa} < a < \cos\frac{j\pi}{\kappa} < b < \cos\frac{(j-1)\pi}{\kappa}, \qquad T_\kappa(a) = T_\kappa(b),$$

one always has

$$\cos\frac{j\pi}{\kappa} - a > b - \cos\frac{j\pi}{\kappa}. \tag{S}$$

Here $\kappa \ge 2$ is even and $1 \le j \le \lfloor \kappa/2 \rfloor$. The original statements appear on source PDF pages 24–25.

with $t' > t$ and so

$$\begin{aligned} X_{(n+3)/2}(\mathcal{E}_n) - \cos(\pi/\kappa) &= X_{(n+3)/2}(\mathcal{E}_n) - \sqrt{\mathcal{E}_n} + \sqrt{\mathcal{E}_n} - \cos(\pi/\kappa) \\ &> t' + \varepsilon > t + \varepsilon \\ &= \sqrt{\mathcal{E}_n} - X_{(n-1)/2}(\mathcal{E}_n) + \varepsilon \\ &> 3\varepsilon. \end{aligned}$$

Again, apply Lemma 6.6 to get $\cos(\pi/\kappa) - X_{(n-3)/2}(\mathcal{E}_n) > 3\varepsilon$. Continuing in this way, we end up with $X_{(n+q)/2}(\mathcal{E}_n) - \cos(\pi/\kappa) > q\varepsilon$ for $q = 1, 3, 5, ..., n$. But $X_{n,n}(\mathcal{E}_n) > n\varepsilon > 1$ is absurd. We conclude that $\ell = \cos^2(\pi/\kappa)$. □

7. A generalization of section 6 : a sequence $\mathcal{E}_n \searrow \cos^2(j\pi/\kappa)$

The construction used to get a sequence in the right-most well of $T_\kappa(x)$ in section 6 is not specific to the right-most well. One can build a similar sequence in other wells of $T_\kappa(x)$.

7.1. **Decreasing sequence in upright well, j odd.** Figure 8 illustrates a decreasing sequence $\mathcal{E}_n \searrow \cos^2(j\pi/\kappa)$, for $\kappa = 8$, $j = 3$. Note that the dotted line $x = \sqrt{\mathcal{E}_n}$ is to the right of the minimum $x = \cos(j\pi/\kappa)$ but converges to it.

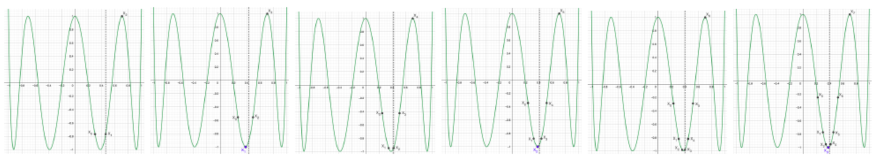

Figure 8. $T_{\kappa=8}(x)$. Solutions $\mathcal{E}_n$. Left to right : $\mathcal{E}_1 \simeq 0.21289$, $\mathcal{E}_2 \simeq 0.18861$, $\mathcal{E}_3 \simeq 0.17584$, $\mathcal{E}_4 \simeq 0.16820$, $\mathcal{E}_5 \simeq 0.16325$, $\mathcal{E}_6 \simeq 0.15983$.

Thus, we propose a generalization of Theorem 1.6 :

Theorem 7.1. *Fix $\kappa \geq 4$, $\kappa \in \mathbb{N}_e$. Fix $1 \leq j \leq \lfloor \kappa/2 \rfloor$, j odd. There is a sequence $\{\mathcal{E}_n\}_{n=1}^\infty$, which depends on κ, s.t. $\{\mathcal{E}_n\} \subset \big(\cos^2(j\pi/\kappa), \cos((j-1)\pi/\kappa) \times \cos(j\pi/\kappa)\big) \cap \mathbf{\Theta}_\kappa(D)$, and $\mathcal{E}_{n+2} < \mathcal{E}_{n+1} < \mathcal{E}_n$, $\forall n \in \mathbb{N}^*$. Also, $\mathcal{E}_{2n-1}$, $\mathcal{E}_{2n} \in \mathbf{\Theta}_{n,\kappa}(D)$, $\forall n \geq 1$, and*

$$\mathcal{E}_n / \cos((j-1)\pi/\kappa) = X_0 < X_1 < ... < X_n < X_{n+1} := \cos((j-1)\pi/\kappa). \tag{7.1}$$

7.2. **Decreasing sequence in upside down well, j even.** For j even, the well is upside down. Figure 9 illustrates a decreasing sequence $\mathcal{E}_n \searrow \cos^2(j\pi/\kappa)$, for $\kappa = 8$, $j = 2$. Note that the dotted line $x = \sqrt{\mathcal{E}_n}$ is to the right of the maximum $x = \cos(j\pi/\kappa)$ but converges to it.

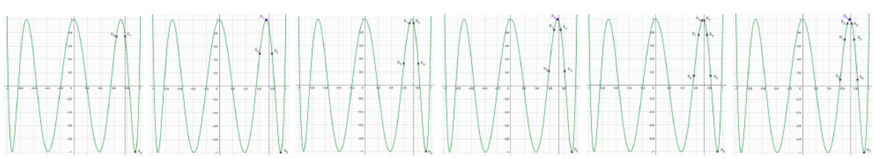

Figure 9. $T_{\kappa=8}(x)$. Solutions $\mathcal{E}_n$. Left to right : $\mathcal{E}_1 \simeq 0.59091$, $\mathcal{E}_2 \simeq 0.56152$, $\mathcal{E}_3 \simeq 0.54484$, $\mathcal{E}_4 \simeq 0.53432$, $\mathcal{E}_5 \simeq 0.52720$, $\mathcal{E}_6 \simeq 0.52212$.

Thus, we propose a generalization of Theorem 1.6 :

Theorem 7.2. *Fix $\kappa \geq 4$, $\kappa \in \mathbb{N}_e$. Fix $2 \leq j \leq \lfloor \kappa/2 \rfloor$, j even. There is a sequence $\{\mathcal{E}_n\}_{n=1}^{\infty}$, which depends on κ, s.t. $\{\mathcal{E}_n\} \subset \left(\cos^2(j\pi/\kappa), \cos((j-1)\pi/\kappa) \times \cos(j\pi/\kappa)\right) \cap \mathbf{\Theta}_\kappa(D)$, and $\mathcal{E}_{n+2} < \mathcal{E}_{n+1} < \mathcal{E}_n$, $\forall n \in \mathbb{N}^*$. Also, $\mathcal{E}_{2n-1}$, $\mathcal{E}_{2n} \in \mathbf{\Theta}_{n,\kappa}(D)$, $\forall n \geq 1$, and*

$$\mathcal{E}_n / \cos((j-1)\pi/\kappa) = X_0 < X_1 < ... < X_n < X_{n+1} := \cos((j-1)\pi/\kappa). \tag{7.2}$$

7.3. **A comment on the proofs of these Theorems and a Conjecture on the limit.** To prove Theorems 7.1 and 7.2 one needs to adapt the proofs of Propositions 6.1, 6.2 and 6.3. The adaptation of these Propositions is straightforward. Moreover, to see why $\{\mathcal{E}_n\} \subset (\cos^2(j\pi/\kappa), \cos((j-1)\pi/\kappa) \times \cos(j\pi/\kappa))$, note that by the construction

$$\cos^2(j\pi/\kappa) < X^2_{(n+1)/2} = \mathcal{E}_n, \quad X_0 \times X_{n+1} = \mathcal{E}_n < \cos(j\pi/\kappa) \times \cos((j-1)\pi/\kappa).$$

As for the limit we conjecture :

Conjecture 7.3. *Let $\{\mathcal{E}_n\}$ be the sequence in Theorems 7.1 and 7.2. Then $\mathcal{E}_n \searrow \cos^2(j\pi/\kappa)$.*

As explained in [GM3] we don't know how to adapt the proof of Lemma 6.6 in order to prove Conjecture 7.3. We do conjecture :

Conjecture 7.4. *Let $\kappa \geq 2$, $\kappa \in \mathbb{N}_e$. Fix $1 \leq j \leq \lfloor \kappa/2 \rfloor$. If $\cos((j+1)\pi/\kappa) < a < \cos(j\pi/\kappa) < b < \cos((j-1)\pi/\kappa)$ are such that $T_\kappa(a) = T_\kappa(b)$, then*

$$\cos(j\pi/\kappa) - a > b - \cos(j\pi/\kappa). \tag{7.3}$$

If Conjecture 7.4 holds, Conjecture 7.3 should follow directly.

8. An increasing sequence of thresholds below $J_3(\kappa) := \left(\cos(2\pi/\kappa), \cos^2(\pi/\kappa)\right)$

This entire section is in dimension 2. We prove the existence of a sequence of threshold energies $\mathcal{F}_n = \mathcal{F}_n(\kappa) \nearrow \inf J_3(\kappa)$. This section proves Theorem 1.10 for $\{\mathcal{F}_n\}$.

Proposition 8.1. *(odd terms) Fix $\kappa \geq 6$, $\kappa \in \mathbb{N}_e$, and $n \in \mathbb{N}_o$. System (5.1) has a unique solution (denoted $\mathcal{F}_n$ instead of $\mathcal{E}_n$) such that $\mathcal{F}_n \in (\cos(\pi/\kappa) \times \cos(2\pi/\kappa), \cos(2\pi/\kappa))$ and*

$$\cos(2\pi/\kappa) =: X_{n+1} < X_n < ... < X_{(n+1)/2} = \sqrt{\mathcal{F}_n} < X_{(n-1)/2} < ... < X_0 < 1, \tag{8.1}$$

*This solution satisfies (5.2), (5.3) and (5.4), and so $\mathcal{F}_n \in \mathfrak{T}_{n,\kappa}$. Furthermore it satisfies (**AO.1**) and so $\mathcal{F}_n \in \mathbf{\Theta}_{(n+1)/2,\kappa}(D[d=2])$.*

Proposition 8.2. *(even terms) Fix $\kappa \geq 6$, $\kappa \in \mathbb{N}_e$, and $n \in \mathbb{N}_e$. System (5.9) has a unique solution (denoted $\mathcal{F}_n$ instead of $\mathcal{E}_n$) such that $\mathcal{F}_n \in (\cos(\pi/\kappa) \times \cos(2\pi/\kappa), \cos(2\pi/\kappa))$ and*

$$\cos(2\pi/\kappa) =: X_{n+1} < X_n < ... < X_{n/2} = \cos(\pi/\kappa) < X_{n/2-1} < ... < X_0 < 1. \tag{8.2}$$

*This solution satisfies (5.10)–(5.13) and so $\mathcal{F}_n \in \mathfrak{T}_{n,\kappa}$. Furthermore it satisfies (**AE.1**) and so $\mathcal{F}_n \in \mathbf{\Theta}_{n/2,\kappa}(D[d=2])$.*

Figure 10 illustrates the solutions in Propositions 8.1 and 8.2 for $1 \leq n \leq 3$ and $\kappa = 6$. Unfortunately the X_q's were accumulating so quickly to $\cos(2\pi/6)$ and 1 that I didn't find a legible way of graphing the cases $n \geq 4$.

2 Proof intuition

The two adjacent branches of T_κ are exactly symmetric in the angular coordinate, not in the horizontal coordinate. Write their points as $a = \cos(\alpha + s)$ and $b = \cos(\alpha - t)$ around the extremal angle $\alpha = j\pi/\kappa$. Equality of the two Chebyshev values forces $s = t$. Converting this angular reflection back to horizontal distances produces the factor $\cos\alpha$. It is positive before the central well, zero at the central well, and negative after it. This gives both the counterexample and the complete corrected statement in one identity.

3 Exact classification

Theorem 1 (Adjacent-branch distance identity)**.** *Let $\kappa \geq 2$ and $1 \leq j \leq \kappa - 1$ be integers. Suppose*

$$\cos\frac{(j+1)\pi}{\kappa} < a < \cos\frac{j\pi}{\kappa} < b < \cos\frac{(j-1)\pi}{\kappa}$$

and $T_\kappa(a) = T_\kappa(b)$. Put $\alpha = j\pi/\kappa$. Then there is a unique $s \in (0, \pi/\kappa)$ such that

$$a = \cos(\alpha + s), \qquad b = \cos(\alpha - s),$$

and

$$\big(\cos\alpha - a\big) - \big(b - \cos\alpha\big) = 2\cos\alpha\,(1 - \cos s). \tag{1}$$

Consequently the source inequality (S) holds for $j < \kappa/2$, is equality for $j = \kappa/2$, and reverses for $j > \kappa/2$.

Proof. Since cosine is strictly decreasing on $(0, \pi)$, there are unique $s, t \in (0, \pi/\kappa)$ such that

$$a = \cos(\alpha + s), \qquad b = \cos(\alpha - t).$$

Using $T_\kappa(\cos\theta) = \cos(\kappa\theta)$ gives

$$T_\kappa(a) = \cos(j\pi + \kappa s) = (-1)^j \cos(\kappa s),$$

and

$$T_\kappa(b) = \cos(j\pi - \kappa t) = (-1)^j \cos(\kappa t).$$

Thus $\cos(\kappa s) = \cos(\kappa t)$. Both arguments lie in $(0, \pi)$, where cosine is injective, so $s = t$. The addition formula now yields

$$\begin{aligned}
&(\cos\alpha - a) - (b - \cos\alpha) \\
&\quad = 2\cos\alpha - \cos(\alpha + s) - \cos(\alpha - s) \\
&\quad = 2\cos\alpha(1 - \cos s).
\end{aligned}$$

Because $s > 0$, the second factor is positive. The three cases follow from the sign of $\cos(j\pi/\kappa)$. ☐

Corollary 1 (Sharp correction of Conjecture 7.4)**.** *For the source range, where κ is even and $1 \leq j \leq \kappa/2$, the assertion is true precisely for $1 \leq j < \kappa/2$. At $j = \kappa/2$ every admissible equal-value pair gives equality rather than strict inequality.*

The smallest explicit counterexample is

$$\kappa = 2, \qquad j = 1, \qquad a = -\frac{1}{2}, \qquad b = \frac{1}{2}.$$

Indeed, $-1 < a < 0 < b < 1$ and, since $T_2(x) = 2x^2 - 1$, $T_2(a) = T_2(b) = -1/2$. But both sides of (S) equal $1/2$.

4 Consequences for Theorems 7.1 and 7.2

The endpoint defect is already present before any convergence argument. At $j = \kappa/2$,

$$\cos^2\frac{j\pi}{\kappa} = 0, \qquad \cos\frac{(j-1)\pi}{\kappa}\cos\frac{j\pi}{\kappa} = 0.$$

Hence the energy interval displayed in both source theorems is $(0, 0)$. No sequence can be contained in this empty interval.

For a separate counterexample to each parity statement, take

$$(\kappa, j) = (6, 3) \quad \text{in Theorem 7.1}, \qquad (\kappa, j) = (4, 2) \quad \text{in Theorem 7.2}.$$

Thus each theorem is false as written. The necessary endpoint correction is $j < \kappa/2$. This packet does not claim that the source's briefly described "straightforward" adaptations prove all remaining instances.

5 Verification, novelty, and limitations

The accompanying SymPy script checks the exact rational counterexample, the trigonometric identity (1), and the two empty-interval examples. It also checks 7,380 adjacent-branch cases for $2 \leq \kappa \leq 41$ on a grid of angular offsets, confirming equality of Chebyshev values and the full sign classification. These computations guard against transcription errors; the argument above proves the infinite statement.

A bounded search on 9 August 2026 used arXiv:2201.00410, the exact title and author, distinctive text from Conjecture 7.4, and combinations of "Chebyshev threshold", "counterexample", and "correction". The arXiv record still contains only v1, and no later correction or exact answer was found. Novelty confidence is moderate pending specialist review.

The result completely settles the elementary adjacent-branch inequality and disproves the endpoint instances of Theorems 7.1–7.2. It does not settle Conjecture 7.3 for the corrected noncentral range, validate the asserted sequence constructions there, or address the paper's broader Mourre and spectral conjectures. Human review should first check the source ranges and then the angular parametrization in the theorem above.

References

[1] M.-A. Mandich, *Thresholds and more bands of A.C. spectrum for the Molchanov–Vainberg Schrödinger operator with a more general long range condition*, arXiv:2201.00410v1, 2022.